

Instructor: Date:

Learner: Regional Branch:

Program: Center:

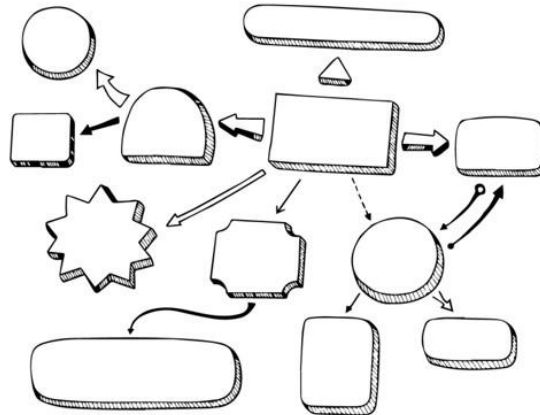
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Let's Start!

Let's organize with diagrams!



Learning Outcome:

Interpret the technical design report, to determine the work plan during the phase of building the software, based on regulation and protocols established by the company.

Design the architecture design using the appropriate tools and types according to the software requirements.

Can Do Statements:

- I can explain what an architecture view model is.
- I can talk about the 4+1 View Model.
- I can describe the four views the 4+1 model provides.

Ask your partner:

- What kind of diagrams do you know?
- What would you use a diagram for in software architecture?





Let's Explore!

Emily is explaining Joshua what a 4+1 model is.



Instructions: Read the text below.

- Joshua:** Hi Emily, what's that diagram you are doing there for?
- Emily:** This? It's just a piping diagram to explain how, where and what the program, I've just designed for the company, is doing.
- Joshua:** Interesting, but I still don't understand plenty of things you've done there, like for example that diagram.
- Emily:** Well, this is just the process review of a whole architecture view model, called 4+1, which allows me to break all the software architecture in multiple views.
- Joshua:** Wow, I think you've said enough. That's a lot of information to work with for me. I'm kind of baffled by what you've said.
- Emily:** Don't worry, I will break it down for you. A 4+1 model allows me to diagram what my program does, how it does it and where it is supposed to work. If you see, the model is divided into 4 different views, the first one is the logical view, as an overall, it describes the object model of the design. The second view is the process view, which describes step by step the relation each activity has within the system, which are a lot by the way. On the other hand, we have the physical view, as you can relate, it describes the mapping of software onto hardware and the distribution aspects, you know, it's like having a computer or a server working only one function of all the software and the relation between that station with the others. Finally, we have the development view that shows what the system looks like from the programmer's perspective, that is, software management.
- Joshua:** And, what is the +1?
- Emily:** Oh right, that's the relation they all have in scenarios where you can test how they are working.





Let's Practice!

Emily is now explaining to his coworkers what an architecture view model is.

Instructions: Read the sentences. Circle the correct word to complete each sentence.

1. (A lot / many / little) of architecture view models represent the whole architectural systems.
2. Remember that (enough / many/ much) architecture view models help the stakeholders to understand the systems from different perspectives.
3. (Much / Many / Plenty) of developers use the architecture view model to see the specific development of software.
4. An architecture view model also allows managers to understand (much / a little / some) processes that a software does.
5. Don't forget that (lots of / few / enough) testers use architecture view models to check the security of a specific software.



Let's Practice!

Emily is giving a report about the implementation of the 4+1 model views in her company. Let's listen to her!

Instructions: Select the correct answer.

1. We _____ the logical view to describe the object model of the design.
 - a. implements
 - b. implementing
 - c. has implemented
 - d. have implemented
2. In our diagram, we also _____ the relation that all the activities had with the system thanks to the process view.
 - a. describes
 - b. described
 - c. describing
 - d. was describing
3. In the physical view of the model, our team _____ the mapping software, hardware, and distribution aspects.
 - a. are represented
 - b. have being
 - c. has represented
 - d. have represented





Let's Practice!

4. Also, the development view _____ us what the system looked like from the programmer's perspective.
 - a. show
 - b. showed
 - c. showing
 - d. are showing

5. So, we _____ these four views to diagram what our programs do, how they do it and where they work.
 - a. has used
 - b. have been
 - c. have used
 - d. is using



Extension Activity

Now, it's time for you to explain the implementation of the 4+1 view model in a company!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Look for more information about the use of the 4+1 view model in a company.
- b) Imagine that you own a software company.
- c) Make a brochure to explain the implementation of the 4 views on your fictitious company.
- d) Share your brochure with your classmates.

Self-Assessment

Yes
No

1. I research about the four views of a 4+1 view model.
2. I create an imaginary software company.
3. I design a brochure to explain the use of the 4+1 model views in a company and I show my ideas to classmates and discuss them.

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Target Words

- **piping diagram** (noun): a diagram that shows the interconnection between processes
e.g. It's just a piping diagram to explain how, where and what the program, I've just designed for the company, is doing.
- **baffled** (adjective): perplex or puzzled about something
e.g. I'm kind of baffled by what you've said.
- **step by step** (idiom): one at a time
e.g. The second view is the process view, which describes step by step the relation each activity has within the system.
- **relate** (verb): to understand a situation
e.g. As you can relate, it describes the mapping of software onto hardware.
- **mapping** (noun): the action of relating and linking objects into one set
e.g. It describes the mapping of software onto hardware and the distribution aspects.
- **station** (noun): a server that is always broadcasting
e.g. It's like having a computer or a server working only one function of all the software and the relation between that station with the others.



Learning Points

Quantifiers

You use **quantifiers** to give information about indefinite quantities.

Quantifiers are always before the object they are modifying. You can use them with countable and uncountable nouns.

Some **quantifiers** that go with countable nouns are *many*, *few*, and *a few*:

Examples:

*There are **many** aspects in that diagram that I can't understand.*
*For the last **few** years, software development has changed enough.*

Some quantifiers that go with uncountable nouns are *much*, *little*, and *a little*.



Learning Points

Examples:

*I think there is not **much** information in that diagram.
There is probably very **little** chance of finishing the whole project in such as short term.*

There are some **quantifiers** that either go before countable or uncountable nouns such as *some, a lot of, lots of, plenty of, and enough*.

Examples:

*I have **plenty of** ideas for our software project.
She is doing **a lot of** effort to finish the presentation.
He thinks he doesn't have **enough** time to prepare the process review.*

Examples

- *I still don't understand plenty of things you've done there.*
- *That's a lot of information to work with.*
- *I think you've said enough.*

What is a view model?

A **view model** is the representation of a system from a specific perspective.

You use **view models** or viewpoint frameworks to represent the whole system in views that will help the stakeholders, end users, developers, managers and testers among others, to understand the system from different perspectives depending on the specific concern, such concern can be security, processes or development.

Among the many **view models**, one of the most important and common ones is the 4+1 view model, it allows the stakeholders to see the system from 4 different views and an extra one that permits to analyze all the 4 views once they are completed.